PROJECT REPORT

Echo Corridors

Bayesian Networks for Path-Based Sensors on Graph-Shaped Environments

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ABSTRACT

This report documents an independent, from-scratch implementation of the path-based sensor Bayesian network of Srivastava, Kontoudis, Sofge and Otte (2026), which fuses boolean "did the agent return" observations from expendable path-following agents into a spatial hazard belief map and plans routes to maximize expected information gain. Rather than reproduce the reference paper's own evaluation environment - an open, cycle-rich 15×15 continuous grid - every experiment here runs on a new environment, Echo Corridors: graph-shaped corridor networks with dead-end rooms and topological bottlenecks, meant to resemble collapsed buildings, tunnels and single-access road networks. That change in environment surfaces two structural gaps in the reference method that its own open-

grid experiments cannot trigger. First, the paper's own prescription for a path that revisits a cell is mathematically under-specified, and the natural reading of it is measurably less accurate than a corrected closed form derived here (log loss against ground truth roughly 3.2× worse over 45 rounds). Second, the multi-robot algorithm assumes a disjoint search-region partition always exists so that per-agent local updates can be merged by a plain set union; on a graph with a genuine cut edge this assumption is false, and the algorithm's specified merge step silently discards one agent's observation of the shared edge, measurably degrading accuracy in the rounds where two agents are forced to share it. A third, cross-cutting observation runs through both: the reference paper's sole convergence metric, Shannon entropy, cannot distinguish a correctly converged belief map from a confidently wrong one, which is exactly why neither gap is visible in its own reported figures. Two further results, on sensor-rate misspecification and on factored versus exact joint beliefs, are reported as tentative. Every figure and number in this document is reproducible from the committed repository.

Contents

1.
Introduction

2.
The Method
2.1
The Bayesian Network Formulation
2.2
The Weighted-Average Baseline
2.3
The Corrected Closed Form for Repeated Visits
2.4
Practical Additions Beyond the Paper

3.
The Echo Corridors Environments
3.1
Random Facility (Single-Agent Revisit Study)
3.2
Bottleneck Facility (Multi-Agent Overlap Study)
3.3
Watching a Round Happen

4.
Implementation
4.1
Module Overview
4.2
Controlling for Confounds

5.
Experimental Setup
5.1
Monte Carlo Parameters
5.2
Metrics

6.
Results

6.1
Revisit Handling
6.2
Multi-Agent Overlap on a Bottleneck
6.3
Entropy Is Not a Correctness Metric
6.4
Sensor-Rate Misspecification
6.5
Factored vs Exact Joint Belief

7.
Findings

8.
Relation to Cited Follow-Up Work

9.
Limitations and Future Scope

10.
Reproducibility

1. Introduction

Changelog. *5 Nov 2026: Experiments 1 and 4 originally reused the initial belief for planning, so the agent repeated one route every round. Fixed; all affected results regenerated. Experiment 5 added.*

A "path-based sensor" produces a single observation for an entire traversed route rather than a measurement at a point. An expendable agent is sent along a path through a hazardous, communication-denied environment; if it returns, the sensor reads 0 (no hazard triggered anywhere on the path); if it does not, the sensor reads 1 (something happened, but not where). Fusing many such observations into a spatial hazard belief map, and choosing routes that gather information fastest, is the problem this report is about.

Srivastava, Kontoudis, Sofge and Otte introduce a Bayesian network formulation of this fusion step, replacing an earlier weighted-average heuristic used in prior work by the same group. The network models hazard existence, per-cell destruction and the resulting sensor trigger as a small directed acyclic graph, and derives closed-form posterior updates for a path that visits each cell at most once. Reported experiments, run on an open 15×15 continuous grid, show faster entropy convergence and fewer destroyed agents than the older method, for both single-agent and parallel multi-agent deployment.

This report documents an independent re-implementation of that method, and evaluates it on a deliberately different environment: graph-shaped corridor networks with dead ends and topological bottlenecks, meant to resemble collapsed buildings, tunnels and single-access road networks rather than open terrain. The choice is not cosmetic. Two of the three confirmed findings below exist only because the environment has a graph topology, and are structurally invisible on a continuous, cycle-rich grid. It contains:

1. a Bayesian-network inference engine implementing three belief-update rules - the reference paper's method, the pre-existing weighted-average baseline it replaces, and a corrected closed form for paths that revisit a cell (`src/inference.py`);
2. a greedy expected-information-gain round-trip path planner over an arbitrary graph (`src/planner.py`);
3. two graph environments - a randomly pruned floor plan with dead ends, and a fixed two-wing facility sharing one bottleneck corridor (`src/environment.py`);
4. a metrics module scoring belief maps against planted ground truth, not only by entropy (`src/metrics.py`);
5. four Monte Carlo experiments and the figures and tables in this report, all regenerated from the committed scripts by `run_all.py`.

Sections 2 through 5 cover the method, the environments, the code, and the experimental setup. Section 6 reports results. Section 7 collects three confirmed gaps and two tentative results into standalone findings, and the remaining sections place this work against the paper's own cited follow-ups and list what it does not attempt.

2. The Method

2.1 The Bayesian Network Formulation

Hazard existence per cell `Z`, per-cell destruction `Delta`, and the path-level sensor trigger `Theta` form a three-layer Bayesian network: an *estimation* layer (`Z`, the belief being maintained), an *inference* layer (`Delta`, whether the agent was destroyed at each cell), and an *observation* layer (`Theta`, the one bit actually seen). Given a survived path (`Theta = 0`), the posterior for a visited cell follows the reference paper's Eq. (7):

$$P(Z_k=1 \mid Delta_k=0) = P(Delta_k=0 \mid Z_k=1) \cdot P(Z_k=1) / P(Delta_k=0)$$

and given a triggered path (`Theta = 1`), the posterior for a given cell follows Eqs. (5)-(6), computed by eliminating the other cells on the path. Both are stated only for a path that visits each cell at most once.

2.2 The Weighted-Average Baseline

The method the reference paper replaces (Otte & Sofge, 2021 [18]; Srivastava et al., 2022 [26]) computes, for a triggered path, the posterior belief map under each mutually exclusive "destroyed at cell `k`" history separately, then combines the `ell` histories weighted by their relative likelihood:

$$P(Z) = \sum_{k=1..ell} P(Z \mid Z_{prior}, Delta_k=1) \cdot P(Delta_k=1)$$

Both this and the Bayesian-network method above are implemented here exactly as published, so that the reported comparisons isolate the belief-update rule itself, not a re-derivation of either.

2.3 The Corrected Closed Form for Repeated Visits

The reference paper's Remark 1 concedes that an agent may revisit a cell, and states only that "the new likelihood `P(Theta|Delta)` is calculated by summing the probability of each instance" - no formula is given, and the claim is not checked against the rest of the derivation. We derive one. Let a round-trip path visit distinct cells `e = 1..m`, with cell `e` visited `n_e` times. Conditioned on hazard state `z_e`, each individual visit is an independent Bernoulli survival trial with probability `a_e = 1 - L` (lethality `L`) if `z_e = 1`, or `b_e = 1 - fp` (false-positive rate `fp`) if `z_e = 0`. Surviving all `n_e` visits to `e` therefore has probability `a_e^n_e` or `b_e^n_e`. Because the agent is destroyed on first failure and does not continue, the observed `Theta` is a deterministic function of whether *any* visit, anywhere on the path, failed:

$$P(\Theta=0 \mid z) = \prod_e \left(a_e^{n_e} \text{ if } z_e=1 \text{ else } b_e^{n_e} \right)$$

Marginalising over the independent priors q_e gives a per-cell survival marginal $s_e = q_e \cdot a_e^{n_e} + (1-q_e) \cdot b_e^{n_e}$, and Bayes' rule gives, for a survived path:

$$P(z_e=1 \mid \Theta=0) = q_e \cdot a_e^{n_e} / s_e \qquad (*)$$

and, for a destroyed path, with $S = \prod_i s_i$ over all distinct cells on the path:

$$P(z_e=1 \mid \Theta=1) = q_e \cdot (1 - a_e^{n_e} \cdot S/s_e) / (1 - S) \qquad (**)$$

Both reduce exactly to the reference paper's Eqs. (7) and (5)-(6) when $n_e = 1$ for every cell. A notable, easily overlooked property of (*) and (**): they depend on the path only through each cell's *visit count*, not the order in which cells were visited - "a failure happened somewhere" is a symmetric function of the underlying independent per-visit outcomes, so a correct implementation needs only to *count* visits correctly, which the paper's own Remark 1 language does not do (Section 6.1 shows what its literal reading produces instead).

2.4 Practical Additions Beyond the Paper

Four details are needed to run the method on a graph and are kept explicit here so a reader can see which parts are the reference method and which are this project's engineering.

Addition	What it does
Round-trip planner	Enumerates "there-and-back" routes (out along a simple path, back the same way - forces a revisit) and "loop" routes (out and back by edge-disjoint simple paths, when the graph offers one), and scores every candidate by exact expected information gain using formula (*)/(**) above.
Shared planning trajectory	In Experiments 1, 2 and 5, one canonical belief map, re-planned from its updated state every round, decides which paths get walked and what happens on them; every compared update rule then replays that identical sequence of observations. This isolates each comparison to the update rule, not to different rules exploring different parts of the map.
`bn_paper_literal`	A best-effort literal reading of Remark 1: a cell visited n times contributes n times its single-visit destruction probability (capped at 1), an additive approximation rather than the correct compounding of survival probabilities in (*)/(**).
Ground-truth scoring	Every belief map is scored against the actual planted hazard map by Brier score, log loss and recall@k, in addition to entropy - see Section 5.2.

3. The Echo Corridors Environments

The reference paper's own experiments use an open 15×15 grid with 8-connected movement in unobstructed 2D space. That choice has two consequences that happen to make both structural gaps in this report invisible. First, revisits never arise: in a grid with no walls, a path that would need to retrace a cell can almost always be rerouted through a neighbouring cell instead, so a round trip - which the paper's own Problem 1 formulation requires, "agents start and end their search rounds at their designated base stations" - is trivial to make revisit-free. Second, disjoint search-region partitions always exist: an entropy-weighted Voronoi diagram over a continuous, simply-connected 2D region can always be drawn so every point belongs to exactly one agent's region, by construction. A graph need not have either property, and surfacing what happens when it does not is the reason this report uses graph environments instead of a grid.

3.1 Random Facility (Single-Agent Revisit Study)

A 4×4 grid of rooms (16 nodes) starts fully connected (up to 24 candidate corridors), then corridors are randomly deleted (deletion probability 0.35) subject to two constraints: the floor plan must stay connected, and at least two dead-end rooms must remain. Across the 40 generated instances used in Section 6.1, the resulting floor plans averaged 16.7 corridors and 4.4 dead-end rooms (range 2-7). Ground-truth hazards are planted independently on each corridor with probability 0.15, matching the reference paper's cell-independence assumption.

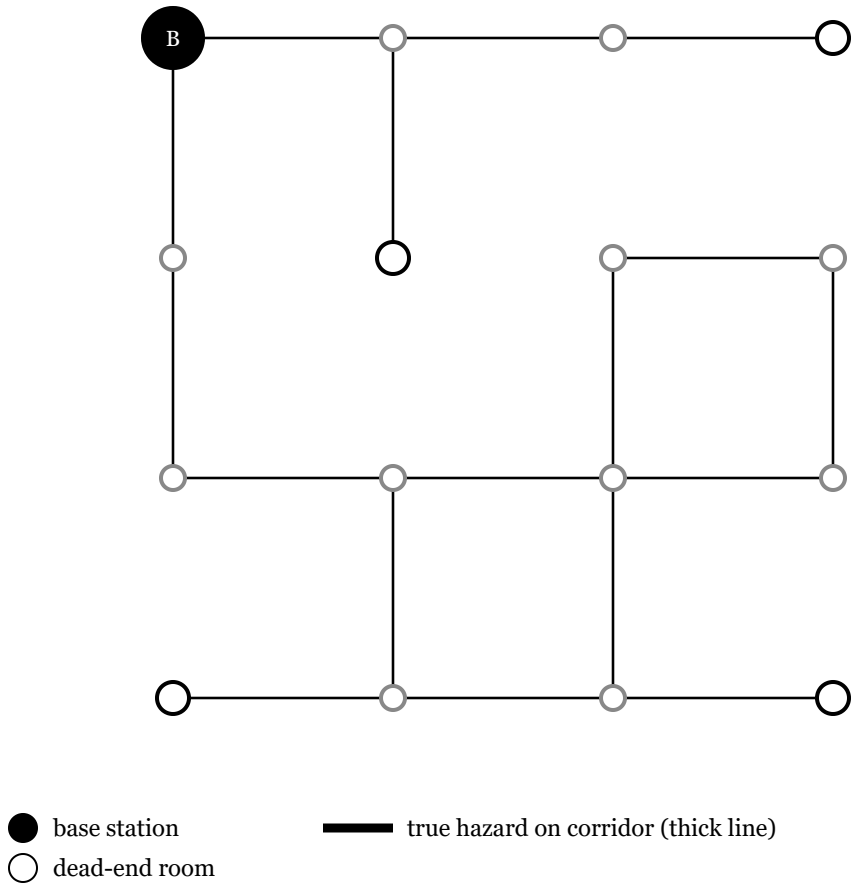


Figure 1. A generated instance. Thin lines are corridors, thick lines carry a planted hazard; the filled node is the base station; open, larger nodes are dead ends the agent can reach only by using one corridor twice on any round trip.

3.2 Bottleneck Facility (Multi-Agent Overlap Study)

A fixed, hand-designed 11-node facility: two independent wings (three rooms each, with their own base stations), a shared junction, one bridge corridor, and a three-room "valley" beyond the bridge that is, every round, the most information-rich unexplored part of the map.

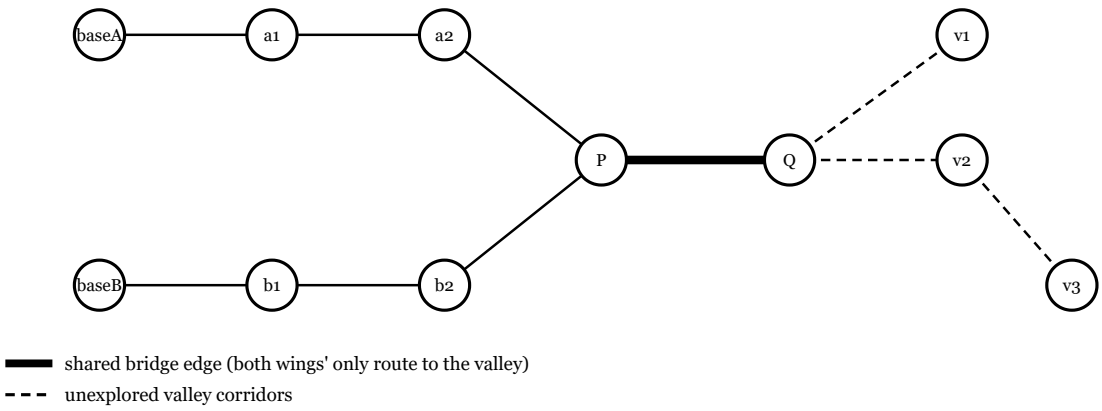
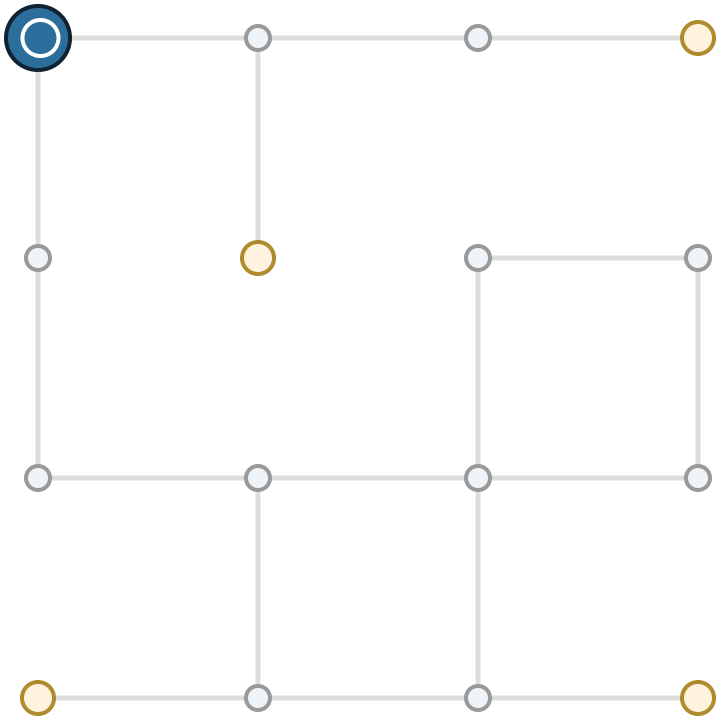


Figure 2. Both wings' only route to the valley is the single marked bridge edge. This is deliberately the simplest possible graph exhibiting the structural problem in Section 6.2: not a rare or contrived layout, but the minimal shape of "two facilities sharing one access route to a place both need to search."

3.3 Watching a Round Happen

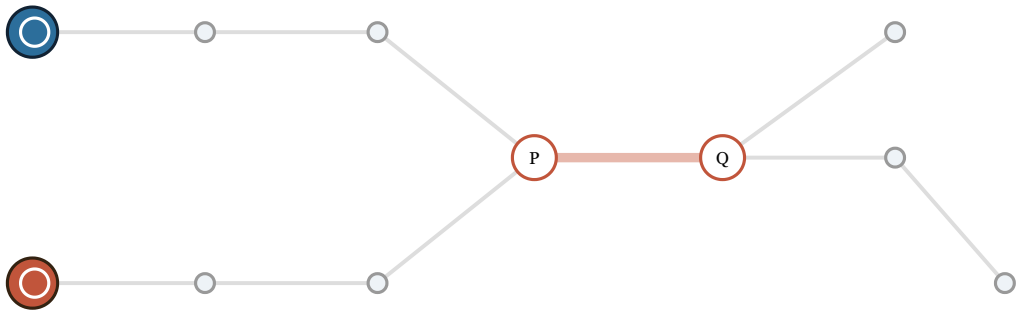
The committed HTML version of this report (`report/echo_corridors_report.html`, opened in a browser rather than printed) embeds two looping SVG animations replaying one representative simulated round each - a colour reproduction of Figures 3 and 4 below with a moving agent marker rather than a single static route. They are SVG rather than GIF because the project has no imaging-library dependency to encode one; see `make_animation.py` for the reasoning. The static routes below are the print-safe equivalent.

One simulated round trip on a random facility (planned round t



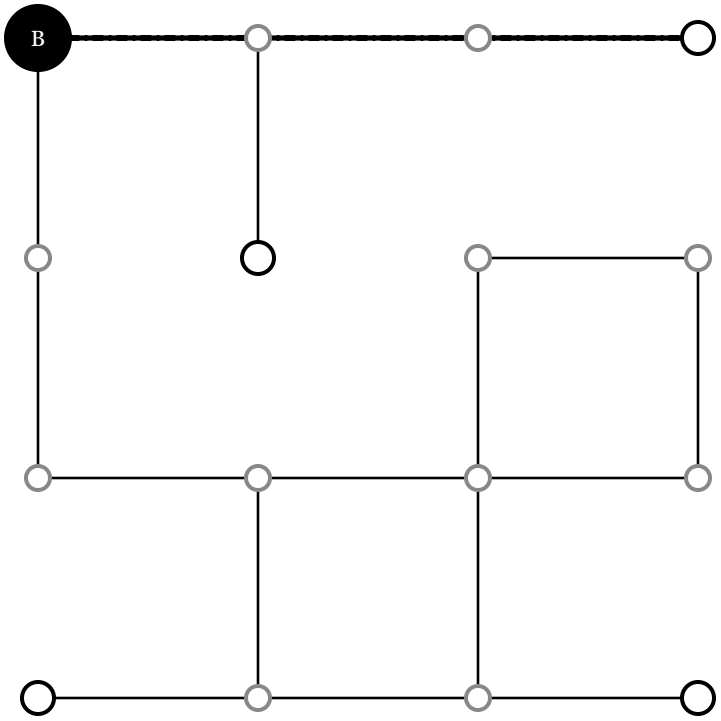
Blue dot = agent's planned round trip from base station B and back (planned round trip revisits a c

Later round, both wings already searched: both agents converge on the shared bridge



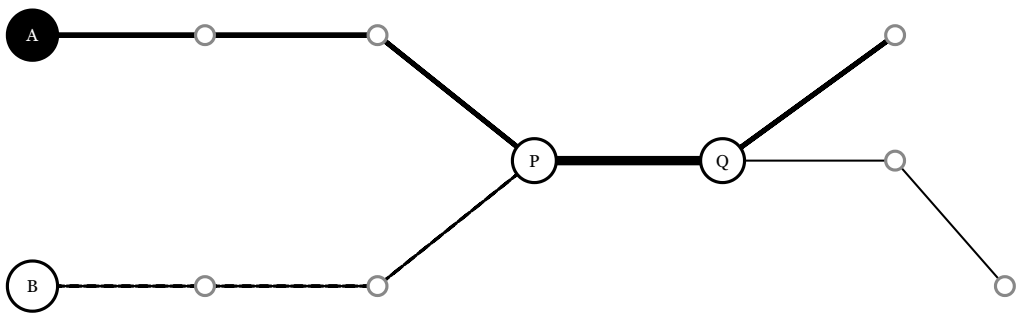
Blue = agent A (wing A), orange = agent B (wing B); both agents' planned routes use the shared bridge edge this round. The thick edge (P-Q)

Live, looping colour replays (visible when this HTML file is opened in a browser; a printed page cannot play them, hence Figures 3-4 below).



Dashed line = agent's planned round trip from base station B and back (planned round trip revisits

Figure 3. One simulated round trip. The dashed route detours into a dead end and is forced to re-traverse the connecting corridor on the way back - the situation Section 6.1 is about.



Solid line = agent A (wing A), dashed line = agent B (wing B); both agents' planned routes use the shared bridge edge this round. The thick e

Figure 4. A later round, after both wings have already searched their own three corridors down to near-certainty: both agents independently plan a route through the shared bridge to the still-unexplored valley in the same round. Solid = agent A, dashed = agent B; where the two lines coincide is exactly the shared edge both must use - the forced-overlap situation Section 6.2 is about.

4. Implementation

4.1 Module Overview

Everything is a from-scratch Python 3 implementation. Experiments 1-4 use only the standard library - no numpy/scipy/matplotlib, so environment, inference and planning logic are all auditable in a few hundred lines each, and the figures in this report are drawn by a small built-in SVG writer rather than a plotting library. Experiment 5 additionally uses numpy, to hold the exact joint posterior over every hazard configuration.

File	Role
`src/environment.py`	Graph environments: random pruned-grid facility generator, fixed bottleneck facility, independent per-edge hazard planting.
`src/inference.py`	Three belief-update rules plus the derivation of the corrected multi-visit closed form (Section 2.3).
`src/planner.py`	Greedy expected-information-gain round-trip path planner: enumerates candidate routes and scores them by exact expected entropy reduction.
`src/metrics.py`	Entropy, Brier score, log loss and recall@k against planted ground truth.
`src/svgplot.py`	Dependency-free SVG line-chart writer used for every chart in this report.
`experiment{1,2,4}_*.py`	Experiments 1, 2 and 4 (Sections 6.1, 6.2, 6.4).
`experiment5_exact_joint.py`	Experiment 5 (Section 6.5): factored vs exact joint belief under two visit models. Requires numpy.
`check_exp5_exact_vs_closed_form.py`	Sanity check: after one update from an independent prior, the exact joint marginals must equal the closed form of Section 2.3.
`run_all.py`	Runs Experiments 1, 2, 4 and 5; writes per-trial raw CSVs, audits every summary mean against them, computes paired differences, runs the sanity check and writes a run manifest.
`make_diagrams.py`, `make_animation.py`	Static and animated figures of both environments, colour and black-and-white variants.

4.2 Controlling for Confounds

Experiments 1, 2 and 5 (Sections 6.1, 6.2, 6.5) hold the *sequence of paths taken and outcomes observed* fixed across the methods being compared - one canonical, best-available belief map, re-planned from its updated state at the start of every round, always drives planning decisions, and every compared update rule is then replayed against that same sequence. This isolates the comparison to the quality of the belief-update rule itself, rather than confounding it with different methods exploring different parts of the map because their beliefs happened to diverge.

5. Experimental Setup

5.1 Monte Carlo Parameters

Experiment	Environment	Trials	Rounds	Lethality	False positive	Hazard prior
1 - revisit handling	Random facility	40	45	0.85	0.04	0.15
2 - bottleneck overlap	Bottleneck facility	400	6	0.85	0.04	0.20
4 - rate misspecification	Random facility	30 / setting	40	0.85 (true)	0.04 (true)	0.15
5 - exact joint belief	Random facility	40 / simulator	45	0.85	0.04	0.15

Seeds are fixed in each script (`20260906` for Experiments 1 and 5, `8675309` for Experiment 2, `42424242` for Experiment 4); re-running `run_all.py` reproduces every number in this report.

5.2 Metrics

The reference paper evaluates convergence with a single metric: total Shannon entropy of the belief map, plus a count of destroyed agents. Entropy measures how *confident* a belief map is, not how *correct* it is - a confidently wrong map has low entropy and looks, by that metric alone, like a success. Alongside entropy, every result below also reports:

- **Brier score** - mean squared error between believed hazard probability and the planted ground truth, averaged over all cells; 0 is perfect.
- **Log loss** - the proper scoring rule most sensitive to overconfident miscalibration.
- **Recall@k** - of the `k` highest-belief cells against the true hazard set, with `k` set to the true hazard count.

6. Results

6.1 Revisit Handling

Three update rules replay the identical sequence of planned paths and observed outcomes each round (Section 4.2): `weighted_average` (the pre-existing method [18]/[26], applied to a revisiting path exactly as its own equations read, since it was never extended for revisits either); `bn_paper_literal` (Section 2.4's best-effort literal reading of Remark 1); and `bn_exact` (the corrected closed form, Section 2.3). Across the 40 trials, agents revisited some corridor an average of 143 extra times per trial over 45 rounds - about three extra visits per round - confirming that in a corridor-network environment, the situation Remark 1 waves at is close to the common case, not a rare edge case.

Table 1. Experiment 1, round 45 of 45, averaged over 40 trials.

Method	Entropy [nats]	Brier score	Log loss	Recall@k
Weighted-average [18]	4.218	0.1012	0.645	0.607
BN, paper's literal revisit rule	3.952	0.0942	0.722	0.534
BN, corrected revisit rule (ours)	4.112	0.0662	0.227	0.655

The corrected rule is the most accurate of the three on every accuracy metric. The literal revisit rule reaches the lowest entropy of the three in every round - it looks the most converged - while having the worst log loss of all three (0.722 vs. 0.227 for the corrected rule, about 3.2×) and worse recall@k than either alternative.

Table 1a. Experiment 1, paired over the same 40 trials, literal minus corrected.

Metric	Difference	SE
Brier score	+0.028	0.0085
Log loss	+0.50	0.13

Positive means the literal rule is worse. SE is the sample standard deviation of the per-trial differences divided by √40.

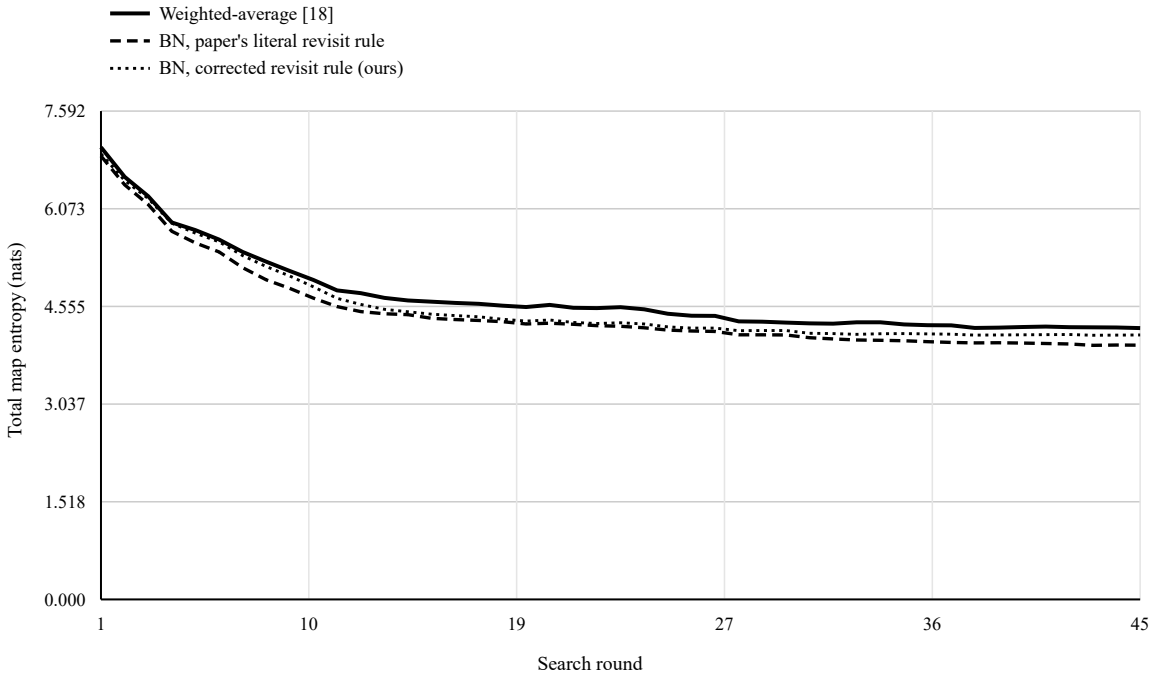


Figure 5. Belief-map entropy vs. search round. The literal revisit rule (dashed) has lower entropy than the corrected rule (dotted) in every round - it looks better converged by the paper's own metric.

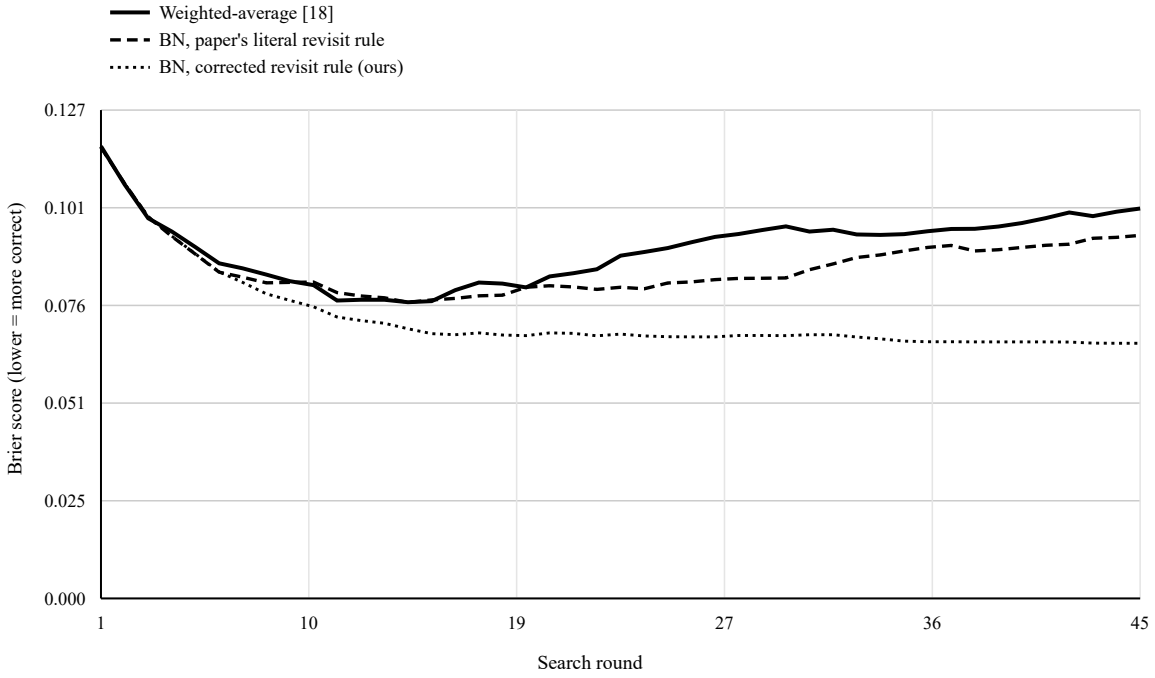


Figure 6. Brier score against the same ground truth. The ranking inverts: the three rules are nearly indistinguishable in the first few rounds, after which the corrected rule (dotted) is the most accurate (lowest in 40 of 45 rounds), with the literal revisit rule (dashed) between it and weighted-averaging

(solid) despite its lower entropy in Figure 5.

A concrete instance: in one trial, an edge with *no* planted hazard was visited twice in a single round trip that ended in the agent's destruction elsewhere on the same path. Starting from the shared prior of 0.15:

Table 2. One representative before/after update (two visits, agent destroyed elsewhere on the same path).

Method	Belief before	Belief after
Weighted-average	0.150	0.374
BN, paper's literal revisit rule	0.150	0.287
BN, corrected revisit rule (ours)	0.150	0.288

At only two visits - the least possible in this environment, since a there-and-back round trip visits every outbound edge exactly twice and no more - the literal and corrected rules are still close. The divergence in Table 1 accumulates from many such two-visit updates recurring on the same frequently used corridors across dozens of rounds, not from any single dramatic event, which is exactly why the paper's own single-round formulation would not have surfaced it.

6.2 Multi-Agent Overlap on a Bottleneck

On the bottleneck facility (Section 3.2), one shared best-available belief map decides both wings' routes each round (Section 4.2), and two ways of folding the round's two independent observations back into the shared map are compared: `naive_union` (exactly Algorithm 3 as specified - each base station computes its local posterior in parallel from the same round-start prior, and the server unions the two local results, discarding one agent's observation of any edge both happened to touch) and `corrected_fusion` (the server instead composes both agents' observations sequentially into the shared prior, which Section 2.3 shows is exact for conditionally-independent evidence about a shared variable). Across 400 trials of 6 rounds each, both wings' planned routes used the shared bridge edge in 18.9% of the 2,400 simulated rounds.

Table 3. Experiment 2, bridge-edge accuracy, 400 trials.

Condition	Naive union	Corrected fusion
Bridge Brier, rounds *with* forced overlap (n=453)	0.088	0.062
Bridge Brier, rounds without overlap (n=1,947)	0.123	0.122
Final-round bridge entropy	0.290	0.259
Final-round bridge Brier score	0.088	0.076
Final-round whole-map Brier score	0.068	0.063

In rounds where the bottleneck was actually shared, discarding one agent's observation left the bridge edge's belief about 30% less accurate than properly fusing both. In rounds without overlap the two strategies are statistically indistinguishable, as expected - the naive union is only wrong precisely when it has to make a choice it should not have to make.

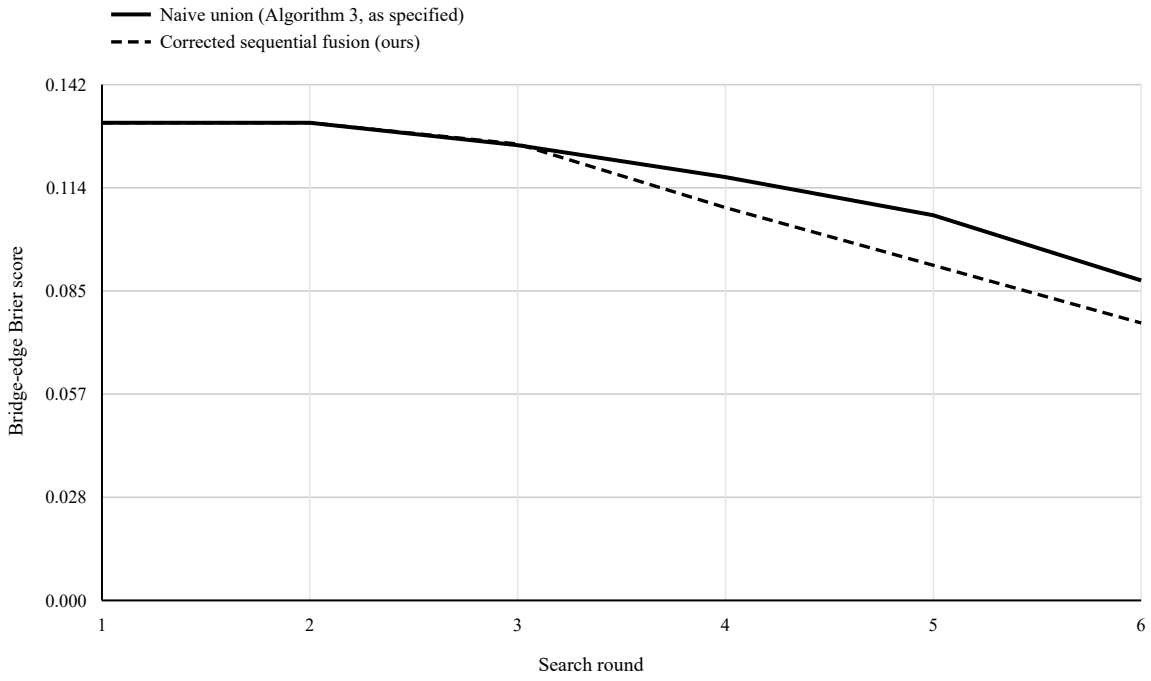


Figure 7. Bridge-edge Brier score vs. search round. Corrected fusion (dashed) stays below naive union (solid) from the first round in which overlap can occur onward.

Algorithm 3's parallel local computation followed by a plain set union is exact if, and only if, the per-round regions are disjoint. The entropy-weighted Voronoi partitioning this builds on constructs regions that are disjoint by construction *in continuous 2D space*. A graph environment with a cut edge has no such guarantee: the bridge edge here is not merely difficult to partition well, it is topologically impossible to assign to one wing without physically cutting the other wing off from the valley. No amount of better Voronoi weighting fixes this, because the problem is not the partitioning heuristic - it is the assumption that a partition can always be found.

6.3 Entropy Is Not a Correctness Metric

This is not a separate experiment; it is a pattern visible in both results above. In Section 6.1, the literal revisit rule reaches lower final entropy (3.952 vs. 4.112 nats, Table 1) than the corrected rule while having about 3.2× worse log loss and a worse Brier score - by the paper's own metric the literal rule looks better converged; by any proper scoring rule against the actual planted hazards, it is the worse one. In Section 6.2, naive union's higher final entropy (0.290 vs. 0.259,

Table 3) happens to agree with its worse Brier score, because that case really is throwing away information rather than manufacturing false certainty. The contrast between the two results is itself informative: entropy can either overstate or roughly track correctness depending on *why* it changed, which is exactly why it cannot be relied on alone to tell the two situations apart without an independent check against ground truth - a check a simulation study can run and a real deployment cannot.

6.4 Sensor-Rate Misspecification

The reference paper tests robustness to false positives and negatives, but in every reported run the belief-update rule is handed the *same* rates used to generate the simulated world. A real deployment estimates these rates from limited field data and is likely to get them at least somewhat wrong, often in the optimistic direction. Holding the true simulated lethality (0.85) and false-positive rate (0.04) fixed, the rate handed to *both* the planner and the belief update was swept away from the truth: a false-positive shrink sweep (assumed rate at 100%, 50%, 25%, 10% and 2% of the true value) and a lethality bias sweep (assumed value biased −35% to +35% relative to the truth), 30 trials per setting, 40 rounds per trial, with the planner re-planning from the Bayesian-network belief every round. Assumed rates are clipped to [0.001, 0.999], so the +20% and +35% lethality levels both clip to 0.999: they are effectively one setting, run with two seed sets.

Table 4. Experiment 4, false-positive shrink sweep, 30 trials/setting.

Assumed false positive	WA Brier	BN Brier	WA log loss	BN log loss	Brier, WA − BN (SE)
100% of true (correct)	0.1006	0.0740	0.451	0.247	+0.0267 (0.0064)
50%	0.0985	0.0761	0.438	0.247	+0.0224 (0.0076)
25%	0.1099	0.0834	0.576	0.266	+0.0265 (0.0077)
10%	0.1214	0.0894	0.764	0.318	+0.0319 (0.0083)
2%	0.1174	0.0793	0.575	0.271	+0.0382 (0.0083)

Table 5. Experiment 4, lethality bias sweep, 30 trials/setting.

Lethality bias (assumed)	WA Brier	BN Brier	WA log loss	BN log loss	Brier, WA − BN (SE)
−35% (0.553)	0.1384	0.0959	0.582	0.299	+0.0424 (0.0086)
−20% (0.680)	0.1113	0.0789	0.597	0.260	+0.0325 (0.0078)
−10% (0.765)	0.1067	0.0732	0.542	0.241	+0.0335 (0.0074)
0% (0.850, correct)	0.0887	0.0653	0.450	0.221	+0.0234 (0.0058)

+10% (0.935)	0.0748	0.0570	0.417	0.211	+0.0178 (0.0056)
+20% (clipped to 0.999)	0.0753	0.0763	0.273	0.284	−0.0010 (0.0019)
+35% (clipped to 0.999)	0.0763	0.0759	0.376	0.347	+0.0004 (0.0033)

WA = weighted average, BN = Bayesian network (corrected rule). The last column is paired over trials; positive means weighted averaging is worse. The two correctly specified rows use different seeds, which is why they differ: run-to-run Brier standard deviation is 0.04-0.11 per setting.

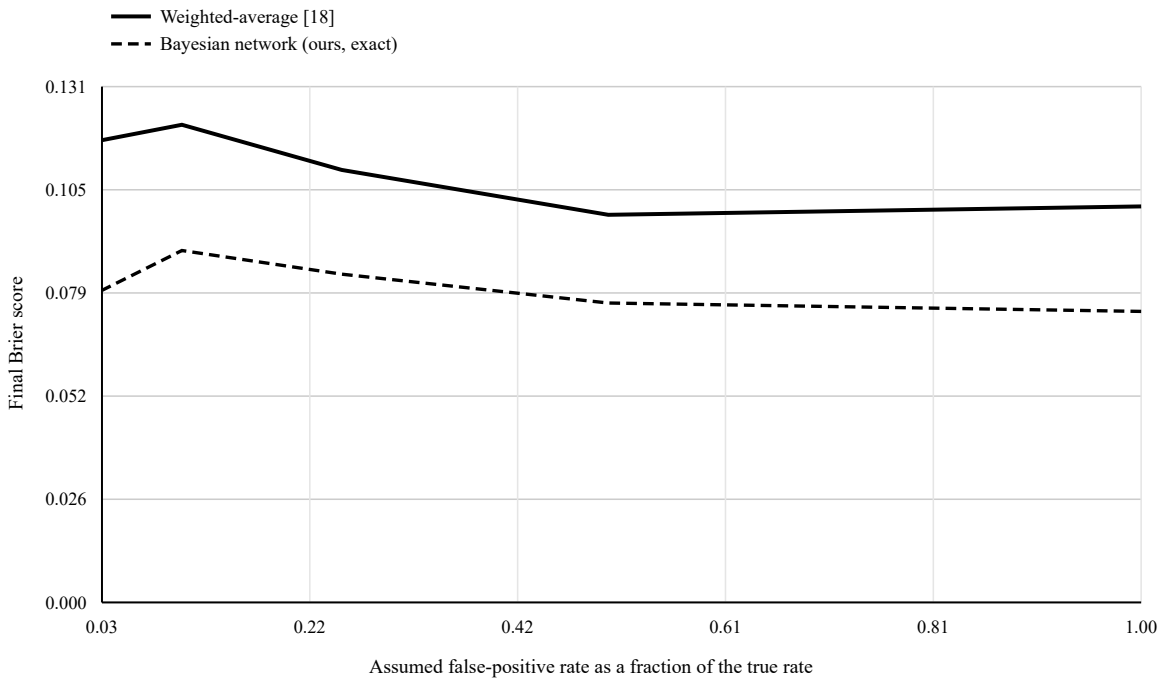


Figure 8. Final Brier score vs. assumed false-positive rate. The Bayesian-network method (dashed) stays below weighted averaging (solid) at every level tested.

The Bayesian-network method beats weighted averaging on both Brier score and log loss at every false-positive level tested, and across lethality bias from −35% to +10%. When lethality is overestimated by 20% or more (the clipped 0.999 setting), the advantage disappears: Brier is about 0.076 for both. The original version of this experiment was reported as inconclusive; that result came from the planning bug noted in the changelog, which made the agent repeat one route every round. The corrected result is still tentative: standard deviations are large, n=30 per setting, the overestimated end is really one clipped setting, and the assumed rate affects both the planner's routes and the belief update at once, so exploration and inference effects are not separated here the way they are in Sections 6.1, 6.2 and 6.5.

6.5 Factored vs Exact Joint Belief

Both the reference paper and this project keep one independent hazard probability per corridor (a *factored* belief). After a "destroyed" observation the corridors on that path become correlated, and storing only per-corridor marginals discards that correlation. Separately, the closed form of Section 2.3 assumes every pass over a corridor is an independent survival trial. Experiment 5 crosses two simulators with four update rules (factored or exact joint, each assuming either visit model):

- **Independent-visit:** every pass over a corridor is a separate survival trial.
- **Once-per-round:** each hazardous corridor gets one destruction draw per round, at the first pass; later passes in the same round add no risk; draws are independent across rounds. (`experiment5_exact_joint.py` calls this model `persistent`.)

Setup matches Experiment 1: the same random facilities and seeds, prior 0.15, lethality 0.85, false-positive rate 0.04, 40 trials and 45 rounds; graphs have 15-19 corridors. The exact joint rule keeps the full posterior over all 2^E hazard configurations (at most 524,288 entries) and updates it with the exact path likelihood. Paths are always chosen by the factored independent-visit belief, re-planned each round, so every rule sees the same paths and outcomes. As a sanity check, the exact joint marginals after one update from an independent prior must equal the closed form of Section 2.3 for both visit models; over 4,000 random cases (`check_exp5_exact_vs_closed_form.py`) the largest difference is 7.7e-15 (independent-visit) and 8.3e-15 (once-per-round).

Table 6. Experiment 5, final round, mean of 40 trials.

Simulator	Rule	Brier	Log loss	Recall@k
independent	factored, independent-visit	0.0662	0.227	0.655
independent	factored, once-per-round	0.0819	0.273	0.640
independent	exact joint, independent-visit	0.0641	0.222	not reported (tie-sensitive)
independent	exact joint, once-per-round	0.0744	0.251	not reported (tie-sensitive)
once-per-round	factored, independent-visit	0.0744	0.342	0.660
once-per-round	factored, once-per-round	0.0751	0.292	0.647
once-per-round	exact joint, independent-visit	0.0720	0.293	not reported (tie-sensitive)
once-per-round	exact joint, once-per-round	0.0644	0.223	not reported (tie-sensitive)

Recall@k is not reported for the exact joint rows because their beliefs near the top-k cutoff differ by less than 1e-9, so floating-point rounding decides the ranking and the value changes between numerical builds.

Table 7. Experiment 5, paired differences over trials (first minus second; positive = first is worse).

Simulator	Comparison	Brier (SE)	Log loss (SE)
independent	factored minus exact (matching model)	+0.0021 (0.0012)	+0.005 (0.003)
independent	wrong visit model minus right (factored)	+0.0157 (0.0041)	+0.046 (0.012)
once-per-round	factored minus exact (matching model)	+0.0106 (0.0041)	+0.070 (0.031)
once-per-round	wrong visit model minus right (factored)	−0.0007 (0.0028)	+0.050 (0.033)

Table 8. How often the planner would pick a different route given exact joint instead of factored marginals (matching visit model).

Simulator	Different route	Mean max marginal gap
independent	22% of rounds	0.077
once-per-round	36% of rounds	0.157

7. Findings

Three properties of the reference method are worth stating on their own, because they are not obvious from the paper or from its own reported experiments and only appear once the method is measured on an environment its own evaluation never used. Two further results are reported as tentative.

FINDING 1 — THE REVISIT RULE IS UNDER-SPECIFIED, AND THE NATURAL READING OF IT IS WRONG

Observation. The reference paper's Remark 1 states that a revisited cell's likelihood is found by "summing the probability of each instance," without giving the resulting formula. Taking that literally and comparing it against the exact closed form derived in Section 2.3, on a floor plan that forces an average of three extra revisits per round, the literal rule ends 45 rounds with about 3.2× worse log loss against ground truth (0.722 vs. 0.227) and a worse Brier score (0.0942 vs. 0.0662), while reaching *lower* entropy - it looks more converged while being less correct.

Why it matters. The paper's formal problem statement requires round trips ("agents start and end their search rounds at their designated base stations"), which is exactly the condition that forces revisits whenever the environment is not fully cyclic. Its own reported experiments use one-way point-to-point traversal on an open grid, which never revisits a cell and therefore never exercises Remark 1 at all. The gap survives publication not because the mathematics is subtle - the correct closed form is a two-line derivation - but because the evaluation environment cannot trigger the code path that needed it.

FINDING 2 — NON-OVERLAPPING PARTITIONS ARE A CONTINUOUS-SPACE ASSUMPTION THAT GRAPHS VIOLATE

Observation. Algorithm 3's per-round server merge is a plain set union of two independently computed local posteriors, valid only if the two agents' regions never overlap. On a facility where both wings' only route to a shared valuable region is one bridge edge, both agents' routes used that edge in 18.9% of 2,400 simulated rounds; the specified union then discards one agent's observation of it, leaving that edge about 30% less accurate than a corrected fusion in the rounds it actually happens.

Why it matters. An entropy-weighted Voronoi partition over continuous 2D space is disjoint by construction. A graph with a cut edge - a single bridge, a single elevator shaft, a single mountain pass - has no such guarantee, and no amount of better Voronoi weighting fixes it, because the problem is not the partitioning heuristic, it is the assumption that a partition can always be found. Any real deployment with a genuine chokepoint inherits this exactly as built here.

FINDING 3 — ENTROPY IS NOT A CORRECTNESS METRIC

Observation. The reference paper's sole convergence metric is Shannon entropy of the belief map. In Finding 1, lower entropy accompanied a worse belief (overconfidence from double-counted evidence); in Finding 2, higher entropy accompanied a worse belief (information genuinely discarded). Entropy moved in opposite directions relative to correctness across the two findings.

Why it matters. A belief map's entropy is a statement about its internal consistency, computable without the true hazard locations. Correctness requires comparing against something the estimator does not have access to in the field - but a simulation study, unlike a real deployment, has the ground truth, and not using it to check for exactly this kind of silent miscalibration leaves failure modes like Finding 1 undetectable by the paper's own reported figures.

FINDING 4 — RATE MISSPECIFICATION: A TENTATIVE, SPECIFIC RESULT

Observation. Feeding the planner and belief update a false-positive or lethality assumption that does not match the true simulated rate, the Bayesian-network method beats weighted averaging on Brier score and log loss at every false-positive level tested (for example, Brier 0.074 vs. 0.101 and log loss 0.247 vs. 0.451 when correctly specified) and across lethality bias from −35% to +10%. When lethality is overestimated by 20% or more, the advantage disappears: Brier is about 0.076 for both. The +20% and +35% levels both clip the assumed lethality to 0.999, so they are effectively one setting run with two seed sets.

Why it matters. The reference paper never tests this axis - its inference procedure always knows its own noise level exactly. This result suggests its resilience claim extends to misestimated rates, except a near-certain assumed lethality. It is tentative: standard deviations are large, n=30 per setting, and exploration and inference effects are not separated, so it needs a larger run before it is relied on.

FINDING 5 — FACTORED VS EXACT JOINT BELIEF, AND THE VISIT MODEL

Observation. With independent hazards, keeping a factored per-corridor belief instead of the exact joint posterior costs little accuracy (Brier +0.0021, SE 0.0012; not clearly significant at n = 40), yet the planner would pick a different route in 22% of rounds; assuming the wrong visit model costs more (Brier +0.0157, SE 0.0041). With once-per-round hazards, the factoring cost roughly quintuples (Brier +0.0106, SE 0.0041) and the planner's route differs in 36% of rounds, while choosing the right visit model inside a factored belief barely helps Brier (−0.0007, SE 0.0028). Recall@k is not reported for the exact joint rules: their beliefs near the top-k cutoff differ by less than 1e-9, so rounding decides the ranking.

Tentative reading. When hazards are independent, decisions are sensitive while outcomes are not, which suggests the information-gain objective is flat across good candidate paths, and the visit model matters more than factoring. When hazards act once per round, the loss of correlations, not the visit assumption, is the larger error. **Limits:** small graphs, one seed family, and n = 40 per cell; the planner always uses the factored independent-visit belief; once-per-round is one of several possible alternatives to independent visits; and the planner's information-gain formula itself assumes a factored belief, even when it is handed exact marginals.

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8. Relation to Cited Follow-Up Work

The reference paper cites several follow-ups by its own authors and others. Based on their published titles and abstracts, none address the three confirmed findings above.

McGuire, Otte & Sofge (2024) [13], "Valuing attrition in a fleet of robots used as path-based sensors," studies how to price the loss of agents against information gained - a decision-theoretic question about when to stop searching, not about whether a given belief update is numerically correct under revisits or overlapping partitions.

Mendelsohn, Sofge & Otte (2024) [14], "Enhancing search and rescue... through path-based sensors with backtracking," studies backtracking as a deliberate *planning* strategy rather than as an unavoidable structural property of corridor topology, and does not, from its abstract, address the Bayesian-network posterior's correctness under repeat visits.

Srivastava, Suresh & Nieto-Granda (2025) [25], "Behaviorally adaptive multi-robot hazard localization in failure-prone, communication-denied environments," addresses adapting agent behaviour to failure-prone conditions - closer in spirit to Finding 4's misspecification question, but framed around behavioural adaptation rather than auditing the inference update's calibration under a wrong assumed fault model, and does not appear to use a graph-topology environment.

None of the cited works, as far as their titles and abstracts indicate, use a graph environment with a genuine topological cut edge, and none formalise or test the multi-visit closed form derived in Section 2.3. The environment design in Section 3 was chosen specifically to expose gaps that a continuous, cycle-rich search space cannot.

9. Limitations and Future Scope

Several choices bound how far these results generalise, and are listed here rather than left implicit.

- **Scale.** All experiments use small graphs (10-19 corridors) so that exact Bayesian inference and exhaustive-enough path enumeration stay tractable; Experiment 5's exact joint posterior grows as 2^E and would not scale to much larger graphs. The mechanism behind each finding is scale-independent, but reported effect sizes should be expected to shift on larger, differently shaped networks.
- **Planner, not the paper's planner.** Path selection here uses a greedy expected-information-gain search over enumerated round-trip candidates, not the reference paper's time-expanded reverse-search algorithm. Both solve the same objective and should behave similarly, but are not code-identical.
- **Weighted-average revisit handling is also ad hoc.** The pre-existing method was never extended for repeat visits either; the baseline here applies it to a revisiting path exactly as its published equations read, which is a defensible "what a practitioner would do," not a claim of uniqueness.
- **Single seed lineage per experiment.** Results are reproducible but drawn from one fixed seed family each (Section 5.1); a wider seed sweep would tighten the confidence bounds, particularly for Findings 4 and 5.
- **Findings 4 and 5 are tentative.** Finding 4 rests on n=30 per setting with large standard deviations, and its overestimated-lethality end is a single clipped setting; Finding 5 rests on n=40 per cell and one choice of alternative visit model.

Two extensions would sharpen this work further: a distributed reconciliation protocol for Algorithm 3 that detects a shared edge before the union step and fuses it correctly rather than requiring a centralised planning belief (used here only to keep the comparison controlled), and a misspecification sweep large enough in trial count, and with more settings above the true lethality, to confirm Section 6.4's tentative result.

10. Reproducibility

Every number and figure in this report comes from the committed repository, at the seeds named in Section 5.1. `python run_all.py` runs Experiments 1, 2, 4 and 5 in sequence, writes per-trial results to `results/raw/`, checks every summary mean against them, writes the paired differences quoted here to `results/paired_differences.json`, runs the Experiment 5 sanity check, and records the git commit, Python and numpy versions, seeds, settings and every output file (with SHA-256) in `results/run_manifest.json`.

- **Tables 1, 1a, 2, Figures 5-6**: `experiment1_revisits.py`.
- **Table 3, Figure 7**: `experiment2_bottleneck.py`.
- **Tables 4-5, Figure 8**: `experiment4_misspecification.py`.
- **Tables 6-8**: `experiment5_exact_joint.py`; sanity check: `check_exp5_exact_vs_closed_form.py`.
- **Figures 1-2**: `python make_diagrams.py`.
- **Figures 3-4** (and the animated HTML versions): `python make_animation.py`.
- All black-and-white print figures: `python make_print_figures.py`, run after `run_all.py`.

Experiments 1, 2 and 4 use only the Python 3 standard library; Experiment 5 also needs numpy. The report itself is `report/echo_corridors_report.html`; the PDF alongside it is that file printed to Letter paper.

Echo Corridors — project report. Method after A. K. Srivastava, G. P. Kontoudis, D. Sofge and M. Otte, Bayesian Networks for Path-Based Sensors: Gathering Information and Path Planning in Communication Denied Environments, 2026, building on M. Otte and D. Sofge, Path-Based Sensors: Paths as Sensors, Bayesian Updates, and Shannon Information Gathering, IEEE Transactions on Automation Science and Engineering, 18(3), 2021, and A. K. Srivastava, G. P. Kontoudis, D. Sofge and M. Otte, Distributed Multi-Robot Information Gathering Using Path-Based Sensors in Entropy-Weighted Voronoi Regions, International Symposium on Distributed Autonomous Robotic Systems, 2022.